🎢 005 · Petting Zoo — Helper Functions — English Worksheet (Beginner)

Topic: Petting Zoo — Helper Functions · **Course:** Theme Park Creator · **Level:** Beginner · **Time:** about 35 minutes

Your park has stalls and amenities. Now it needs a **petting zoo** — a fence around it, six cages inside, an animal in every cage, and somewhere to eat and drink.

Every cage is the same job. So you write that job **once**, then **call** it for every animal.

These are the tools you use this lesson:

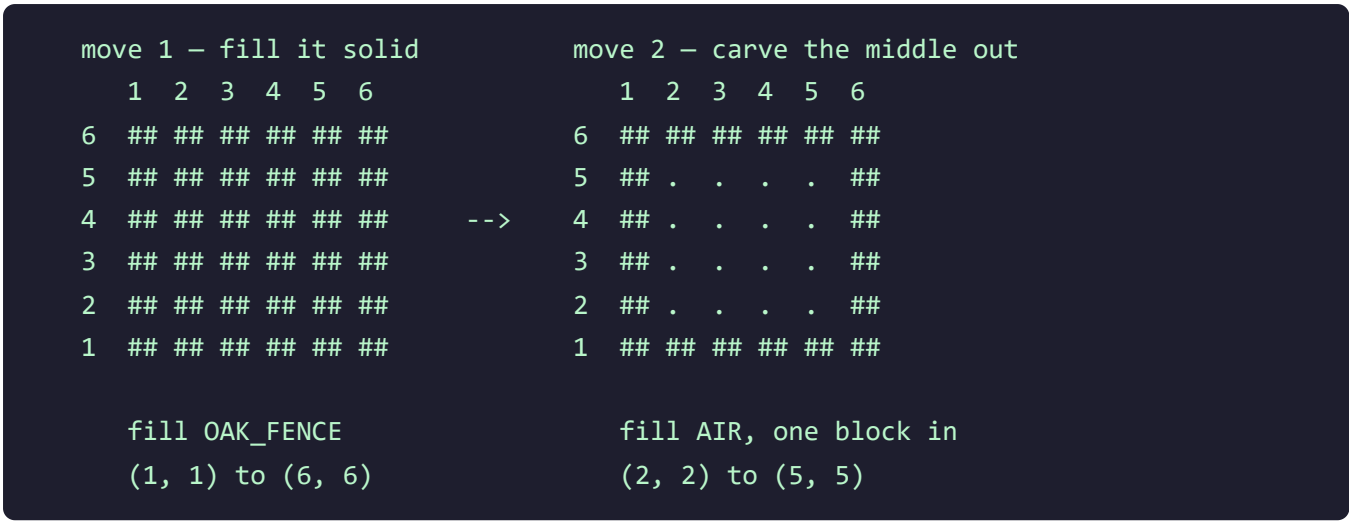
```
corner1 = pos(1, 0, 1)
corner2 = pos(10, 0, 10)
blocks.fill(OAK_FENCE, corner1, corner2, FillOperation.REPLACE)
mobs.spawn(CHICKEN, pos(6, 0, 6))

for i in range(10):
    mobs.spawn(CHICKEN, pos(6, 0, 6))
```

> `corner1` and `corner2` are the **two corners** of the box you fill. `FillOperation.REPLACE` fills the **whole solid** box. `AIR` is a block like any other — filling with `AIR` **deletes** what was there. `mobs.spawn(animal, pos(x, y, z))` puts **one** animal in **one** spot. `for i in range(10):` runs the line under it **10 times**. `OAK_FENCE` = the fence. `HAY_BLOCK` = the food. `WATER` = the drink.

These are the tools. **Turning the numbers into parameters is your job.**

1 · Solid, Then Carve 🧱

There is no "hollow box" block. So you build a box in **two moves**:



Move 2 is one block inside move 1 on every side. The edge it leaves behind is your fence.

2 · Functions 🧩

What a helper function is

	Regular function	Helper function
Who calls it?	**You** do. You type it and run it.	**Another function** does.
How big is the job?	The whole thing.	**One** small job.
In your zoo	`build_zoo`	`fence` · `feed_area`
Written differently?	No — same `def`.	No — same `def`.

> ⚠️ There is **no special keyword** for a helper. A helper is just a function that **other functions use**. The word describes its **job**, not its code.

How to tell them apart: look at who calls it. If you type it yourself, it's the regular one. If only other functions type it, it's a helper.

Functions inside functions 🪆

A function can **call** another function. When it does, it **stops and waits** — the second function runs all the way to the end, then the first one carries on from where it stopped.

Follow the arrows. `-->` means *go into*. `<--` means *finished, come back*.

```
YOU TYPE:  build_zoo()
     |
     |  line 1 --> build_cage(CHICKEN, 1, 1, 10, 10)
     |                  |
     |                  |-- fill the fence box
     |                  |-- carve the middle out
     |                  |-- spawn the chicken
     |             <-- finished. back to build_zoo.
     |
     |  line 2 --> build_cage(COW, 13, 1, 22, 10)
     |                  |-- same three jobs, new numbers
     |             <-- finished. back to build_zoo.
     |
     |  line 3 --> build_cage(PIG, 25, 1, 34, 10)
     |             <-- finished. back to build_zoo.
     |
     v  (three more, then build_zoo is done)
```

The rule: nothing is skipped. `build_zoo` cannot get to its next line until the function it called has completely finished.

Work it out step by step:

Step 1. You type `build_zoo()`. Only that. One line.

Step 2. `build_zoo` runs its **first** line. That line calls `build_cage`.

Step 3. `build_zoo` **stops and waits**. `build_cage` is running now.

Step 4. `build_cage` does its three jobs — fence, carve, animal. Then it ends.

Step 5. `build_zoo` **wakes up** and runs its **second** line. Another `build_cage`.

Step 6. Repeat until all six are done.

How to write a helper 🧰

Four steps, every time:

Step 1 — Find the job that repeats. Six cages. Same job, six times.

Step 2 — Write the job once. Give it a name that says the job.

Step 3 — Put the changing parts in the brackets. Those become **parameters**.

Step 4 — Call it, with different arguments each time.

Here is the **plan** for `build_cage`. The code is yours to write — this is only the order of the jobs:

```
def build_cage(animal, x, z, x2, z2):
    # 1. fill a solid box of fence, corner to corner
    # 2. carve the middle back to AIR, one block in
    # 3. put the animal inside
```

3 · Predict 🔮

Read the code. Circle your answer.

```
blocks.fill(OAK_FENCE, pos(1, 0, 1), pos(10, 0, 10),
    FillOperation.REPLACE)
```

What is on the ground now? Circle one: a solid square of fence · a fence ring · nothing

```
blocks.fill(AIR, pos(2, 0, 2), pos(9, 0, 9), FillOperation.REPLACE)
```

Now what is left? Circle one: a fence ring · a solid square · nothing

The animal has to stand inside. Which move makes room for it? Circle one: move 1 · move 2

In `build_cage(SHEEP, 1, 13, 10, 22)`, what goes into the box named `animal`? Circle one: `SHEEP` · `1` · `13`

4 · Spot the Bug 🐛

Bug A — This should leave a fence ring with a space inside.

```
blocks.fill(OAK_FENCE, pos(1, 0, 1), pos(10, 0, 10),
    FillOperation.REPLACE)
blocks.fill(AIR, pos(1, 0, 1), pos(10, 0, 10),
    FillOperation.REPLACE)
```

What is left on the ground? Circle one: nothing · a fence ring · a solid square

Why does that happen?

Bug B — Six animals should stand in six different cages.

```
build_cage(CHICKEN, 1, 1, 10, 10)
build_cage(COW, 1, 1, 10, 10)
build_cage(PIG, 1, 1, 10, 10)
```

What is wrong? Circle one: every cage is in the same spot · the animals are the same · the helper is missing

How do you know?

5 · Build the Petting Zoo 🐑

Switch to your homework world. **Work through these in order — do not skip to step 6.**

Step 1. Write the line `def build_cage(animal, x, z, x2, z2):`

Step 2. Inside it, make `corner1` and `corner2` out of the **parameters** — not out of fixed numbers.

Step 3. Fill that box solid with `OAK_FENCE`.

Step 4. Fill `AIR` **one block inside** the box. Work out what to add to `x` and `z`, and what to take off `x2` and `z2`.

Step 5. Spawn the animal **in the middle**, not on the corner.

Step 6. Call it **once**, with `CHICKEN`. Run it. Look at it. **Fix it before you go on.**

Step 7. Now write the other five calls, with five different animals.

Step 8. Put one big `OAK_FENCE` box around the whole zoo.

Animals: `CHICKEN` · `COW` · `PIG` · `SHEEP` · `RABBIT` · `LLAMA`

Cage corners that fit — front row and back row:

```
z 13-22    [ SHEEP  ]   [ RABBIT ]   [ LLAMA  ]

z  1-10    [ CHICKEN]   [  COW   ]   [  PIG   ]

            x 1-10       x 13-22      x 25-34
```

Write your helper and your six calls:

Your helper is written once but runs six times. Explain how that works.

6 · Show Your Work 📸 🎥

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Say these out loud, filling in the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.
>
> The most fun part was _____.
>
> Something new I learned was _____.

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